

CLIQUE: Grid-Based Subspace Clustering

- ❑ CLIQUE (Clustering In QUES) (Agrawal, Gehrke, Gunopulos, Raghavan: SIGMOD'98)
- ❑ CLIQUE is a density-based and grid-based subspace clustering algorithm
 - ❑ Grid-based: It discretizes the data space through a grid and estimates the density by counting the number of points in a grid cell
 - ❑ Density-based: A cluster is a maximal set of connected dense units in a subspace
 - ❑ A unit is dense if the fraction of total data points contained in the unit exceeds the input model parameter
 - ❑ Subspace clustering: A subspace cluster is a set of neighboring dense cells in an arbitrary subspace. It also discovers some minimal descriptions of the clusters
- ❑ It automatically identifies subspaces of a high dimensional data space that allow better clustering than original space using the Apriori principle

CLIQUE: SubSpace Clustering with Apriori Pruning

- ❑ Start at 1-D space and discretize numerical intervals in each axis into grid
- ❑ Find dense regions (clusters) in each subspace and generate their minimal descriptions
 - ❑ Use the dense regions to find promising candidates in 2-D space based on the Apriori principle
 - ❑ Repeat the above in level-wise manner in higher dimensional subspaces

CLIQUE: SubSpace Clustering with Apriori Pruning



Major Steps of the CLIQUE Algorithm

- ❑ Identify subspaces that contain clusters
 - ❑ Partition the data space and find the number of points that lie inside each cell of the partition
 - ❑ Identify the subspaces that contain clusters using the Apriori principle
- ❑ Identify clusters
 - ❑ Determine dense units in all subspaces of interests
 - ❑ Determine connected dense units in all subspaces of interests
- ❑ Generate minimal descriptions for the clusters
 - ❑ Determine maximal regions that cover a cluster of connected dense units for each cluster
 - ❑ Determine minimal cover for each cluster

Pros and Cons of CLIQUE

☐ Strengths

- ☐ Automatically finds subspaces of the highest dimensionality as long as high density clusters exist in those subspaces
- ☐ Insensitive to the order of records in input and does not presume some canonical data distribution
- ☐ Scales linearly with the size of input and has good scalability as the number of dimensions in the data increases

☐ Weaknesses

- ☐ As in all grid-based clustering approaches, the quality of the results crucially depends on the appropriate choice of the number and width of the partitions and grid cells